Tine
Starič

BC/NAV TECH DAYS 2024

mibuso.com

Reviewing the Code Review

BC/NAV TECH DAYS 2024
mibuso.com

Tine Starič

Software Architect @ Companial

Microsoft® MVP Most Valuable Professional

44 : 07 : 26 : 48

DAYS HOURS MINUTES SECONDS

Agenda

- Why do we do code review
- Review the process
- Review the reviewer
- Review the reviewee
- Are Copilots going to help?

Why Code Review?

- Ensuring code quality
- Following coding guidelines
- Find bugs and edge cases

Why Code Review

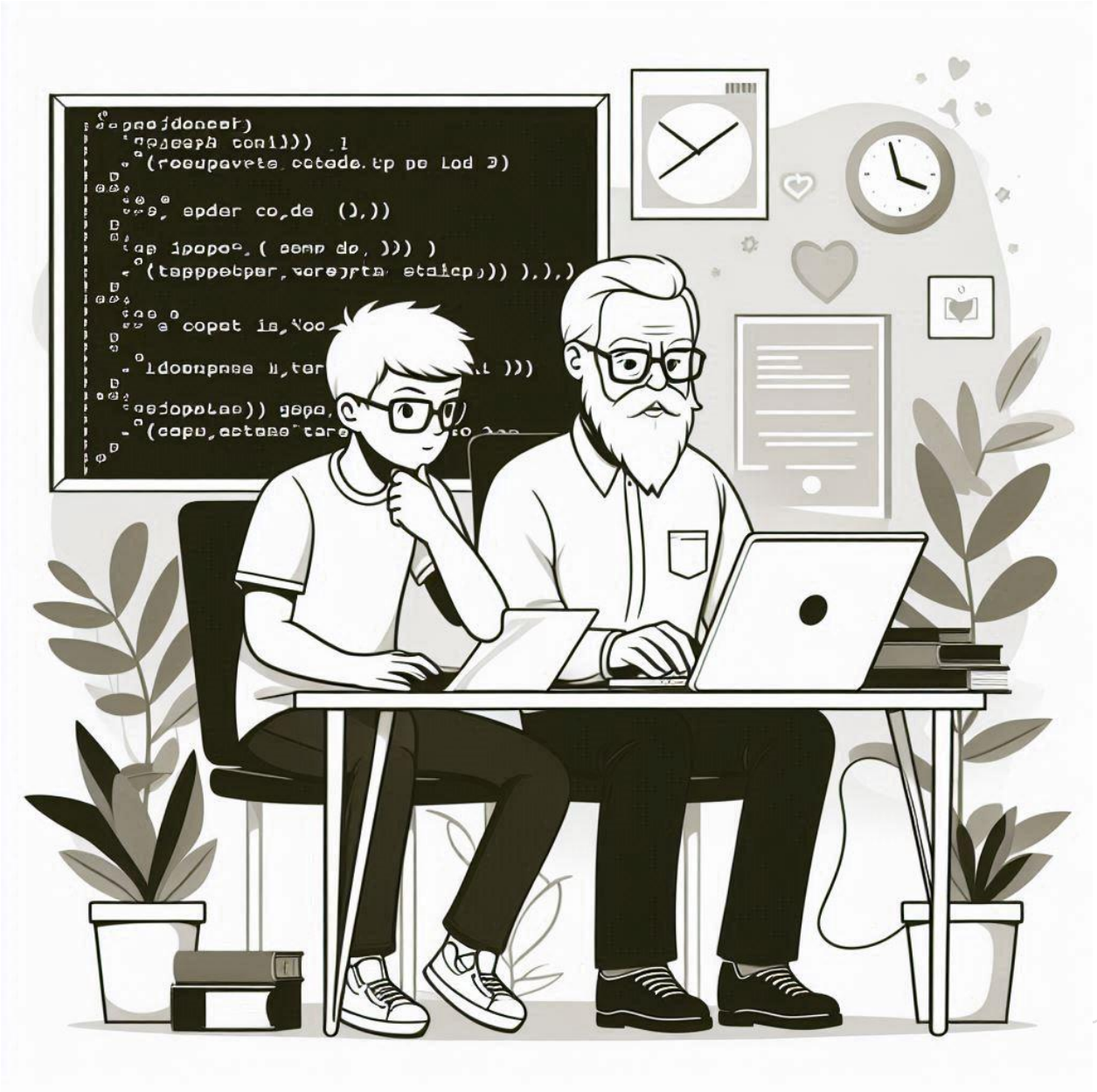
“Ask yourself "If I had to support this in production tomorrow, with no previous knowledge of it, how tempted would I be to send a glitter bomb to the developer who wrote it?"

When you think like that, it really cleans up a lot of your code”

– Unknown Redditor

Avoiding the lottery factor of 1

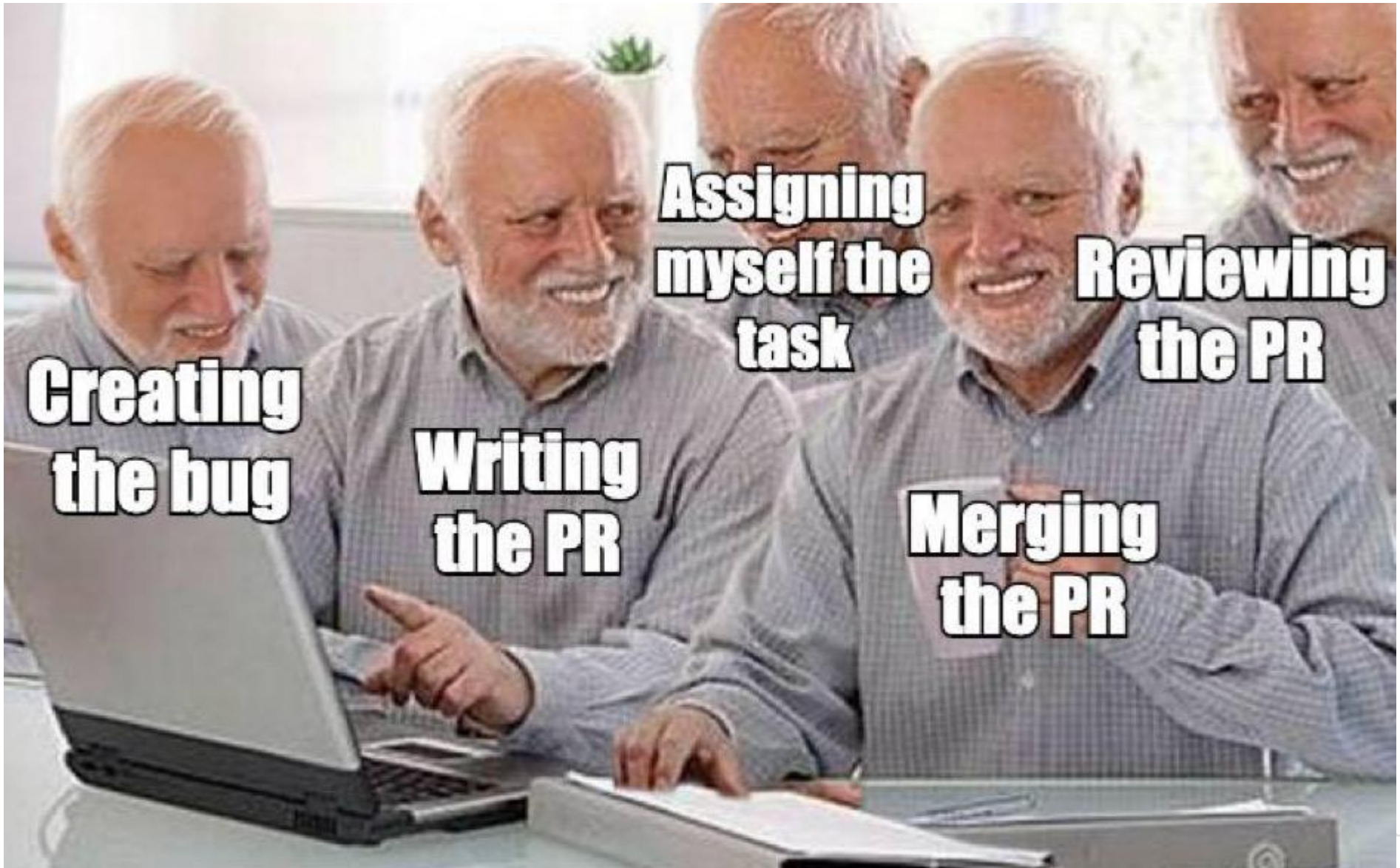
Knowledge Transfer



Why Code Review?

- Ensuring code quality
- Following coding guidelines
- Find bugs and edge cases
- Avoiding the lottery factor of 1
- Knowledge transfer

Why Code Review?



Review the process

Pull Requests, policies, pipelines and linters

Review the process

Pull Requests, policies, pipelines and linters

- Does it build?
- Do all the tests pass?
- Are translations done?
- Is it following the coding guidelines?

Linting use cases – Coding Conventions

BC/NAV TECH DAYS 2024
mibuso.com

```
You, 54 seconds ago | 1 author (You)
var
    2 references
    HideMessage: Boolean;
```

```
(global) HideMessage: Boolean
Global variables should be prefixed with "g" AL(TS0002)
View Problem (Alt+F8)    Quick Fix... (Ctrl+.)
```

```
if SkipCheck then
    exit;
```

```
The exit keyword ends the procedure with the specified exit value.
EXIT keyword must be Capitalized AL(TS0005)
View Problem (Alt+F8)    Quick Fix... (Ctrl+.)
```

Linting use cases – Maintainability improvements

BC/NAV TECH DAYS 2024
mibuso.com

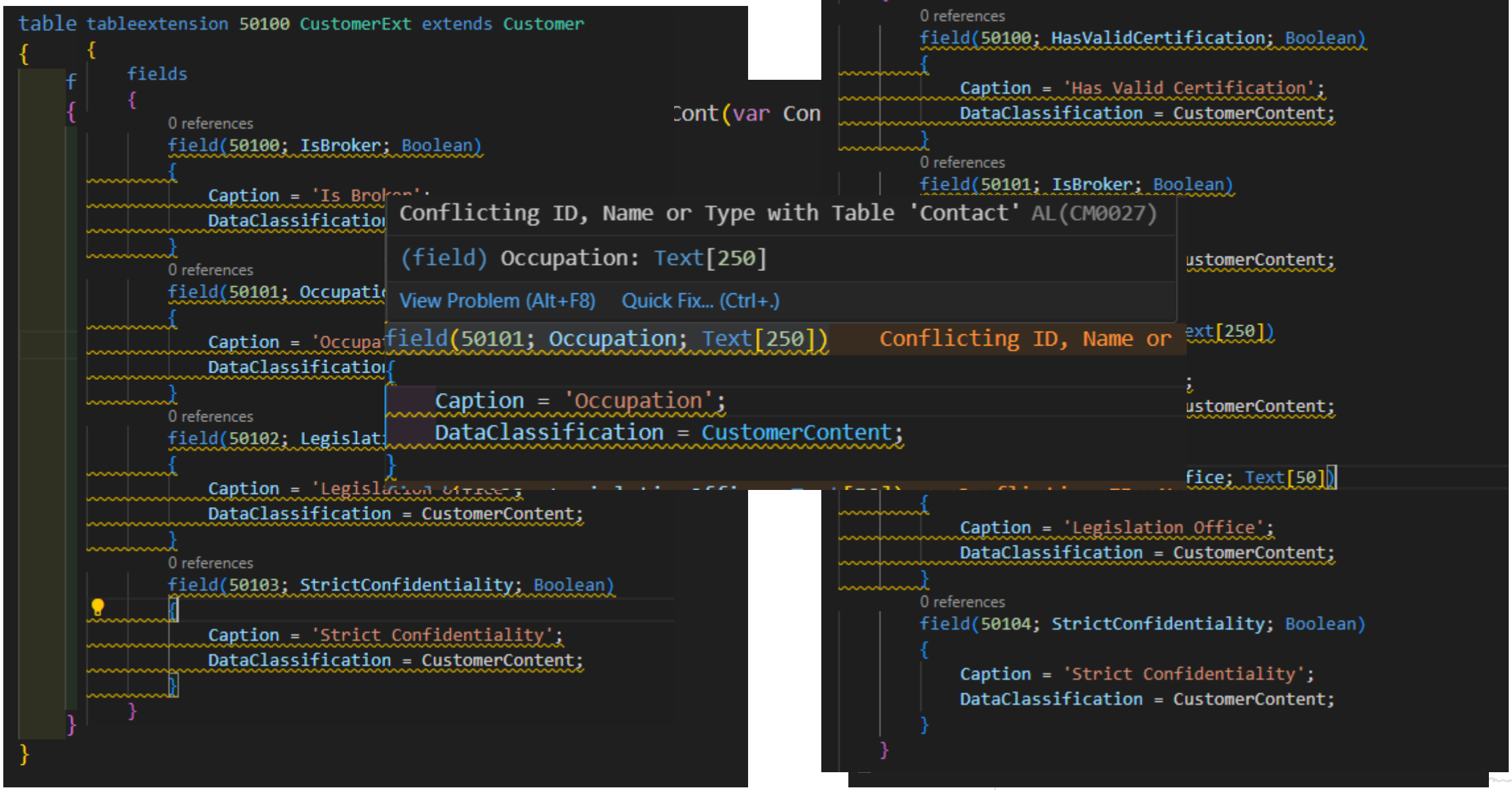
```
1 reference | 0% Coverage
local procedure ImportVendorBankAccounts()
begin
    ImportBankAccountsFromExcel();

    Commit();

    AssignPrimaryBankAccount();
end;
```

```
1 reference | 0% Coverage
local procedure CreateNewCustomer(Number: Code[20]; Name: Text[100])
var
    Customer: Record Customer;
begin
    Customer.Init();
    Customer."No." := Number;
    Customer.Name := Name;
    Customer.Insert();
end;
```

```
procedure Commit()
Ends the current write transaction.
Commit() needs a comment to justify its existance. Either a leading or a trailing comment. AL(TS0012)
View Problem (Alt+F8)    Quick Fix... (Ctrl+.)
```

```
procedure Insert(): Boolean
Inserts a record into a table without executing the code in the OnInsert trigger.
Internal Methods must be invoked with explicit parameters AL(TS0013)
View Problem (Alt+F8)    Quick Fix... (Ctrl+.)
```

Linting use cases



BC/NAV TECH DAYS 2024
mibuso.com

Review the process

Pull Requests, policies, pipelines and linters

- Does it build?
- Do all the tests pass?
- Are translations done?
- Is it following the coding guidelines?

Review the process

Size, time and frequency of pull requests

- Many smaller pull requests are better than one huge
- Make small pull requests
- No, seriously, keep pull requests small

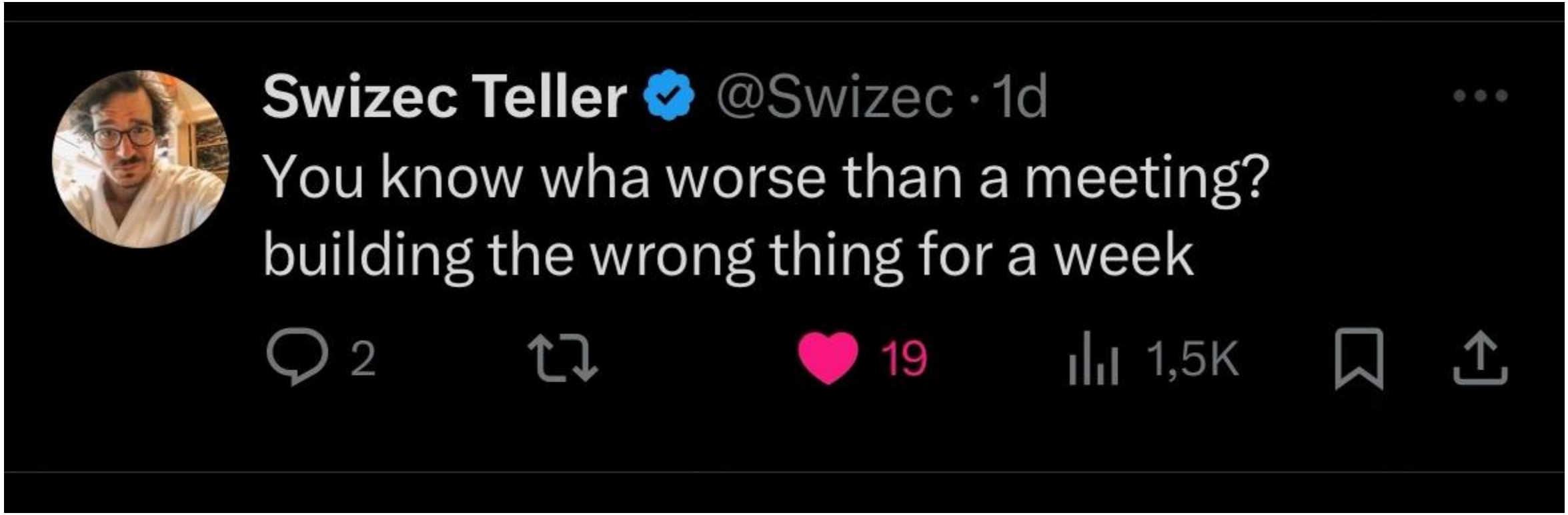
Small work item – Small Pull Request

"But when the code's finished is too late for deep feedback. Any pushback breeds resentment. What do you mean you don't like my beautiful baby!?"

– Swizec Teller

https://swizec.com/

Small work item – Small Pull Request



Review the process

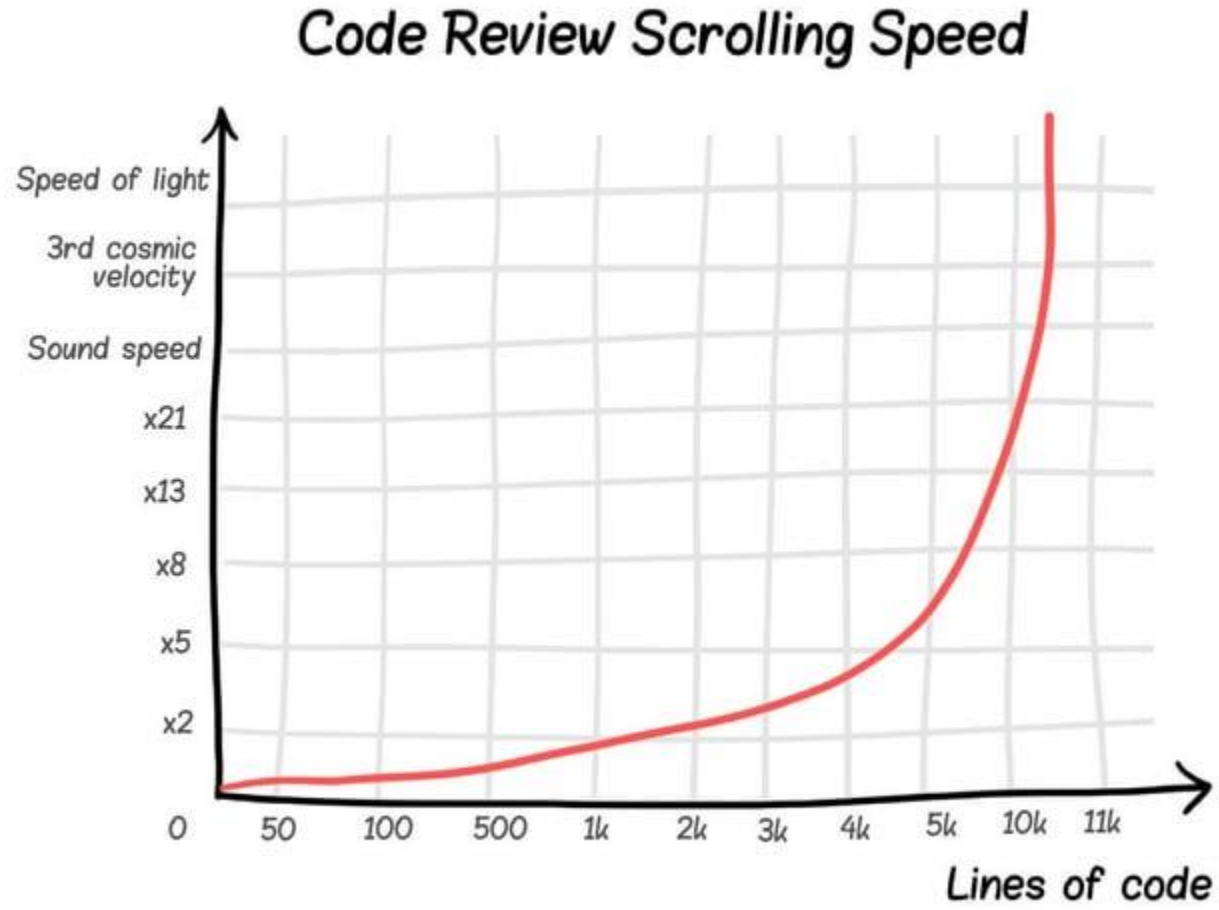
Size, time and frequency of pull requests

- Many smaller pull requests are better than one huge
- Make small pull requests
- No, seriously, keep pull requests small

- Small work items will result in a small PR.
- Ideal work span is up to 3 days, 5 is okay
- 1 hour is too little, 2 weeks are questionable, more than two weeks should never happen.

- If you're doing big PRs, plan time for code review

Make it small!



Make it small!

Ryan Peterman
@ryanlpeterman

Ask an engineer to review a 15 lines of code and they'll find plenty of issues.

Ask them to review 1000 lines and they'll say it looks good.

44 Retweets **661** Likes

Reviewing the Reviewer

- Be timely – Prioritize review work to unblock others
- Review the changes not just the syntax
- Stick to the scope

Review

Coding guide

- Different
- The ultim

https://algu

Code formatting

Code should follow the AL formatting rules enforced by VS Code. No individual formatting rules should be set because different formatting ends up with unnecessary changes in each commit.

In general, there is no restriction on the line length, but lengthy lines can make the code unreadable. We recommend having a max length of 120 symbols to keep code easily scannable and readable.

When considering the issue of excessively long procedure signatures or invocations, it becomes imperative to maintain code readability and conformity to established coding standards. In cases where a procedure signature or invocation spans an excessive length, it is recommended to break the parameters into separate lines to enhance code clarity. Proper indentation plays a pivotal role in achieving this objective. An exemplary illustration of correct indentation involves aligning the parameters in a manner that facilitates quick comprehension, typically by placing each parameter on a new line and indenting them appropriately, as demonstrated below:

```
internal procedure SetAddress(
    Address: Text[100];
    Address2: Text[50];
    PostCode: Code[20];
    City: Text[30];
    County: Text[30];
    CountryCode: Code[10];
    Contact: Text[100]
)
begin
end;
```

```
trigger OnRun()
begin
    SetAddress(
        Address,
        Address2,
        PostCode,
        City,
        County,
        CountryCode,
        Contact
    )
end;
```

A Wild West of coding

BC/NAV TECH DAYS 2024
mibuso.com

```
[Scope('OnPrem')]
9 references | 0% Coverage
procedure VoorraadboekingsgroepGebruiken(f_Voorraadgroep: Code[20]): Boolean
var
    VoorraadgroepenPerBedrijfLrec: Record "Voorraadgroepen per bedrijf";
    lText001: Label 'U heeft geen recht om dit artikel met verkoopboekingsgroep: %1 te gebruiken in deze vestiging!';
begin
    //INDEX 3
    if VoorraadgroepenPerBedrijfLrec.FindFirst then // controle bestand gevuld en functionaliteit wordt gebruikt
        if f_Voorraadgroep <> '' then
            exit(VoorraadgroepenPerBedrijfLrec.Get(f_Voorraadgroep))
        else
            exit(false);

    exit(true);
end;
```

How will you treat “this”?

BC/NAV TECH DAYS 2024
mibuso.com

```
codeunit 50100 WhatIsThis implements ISomethingInteresting
{
    0 references
    trigger OnRun()
    var
        What: Codeunit What;
    begin
        What.Is(this);
        this.IsComingSoon();
    end;
```

Reviewing the Reviewer

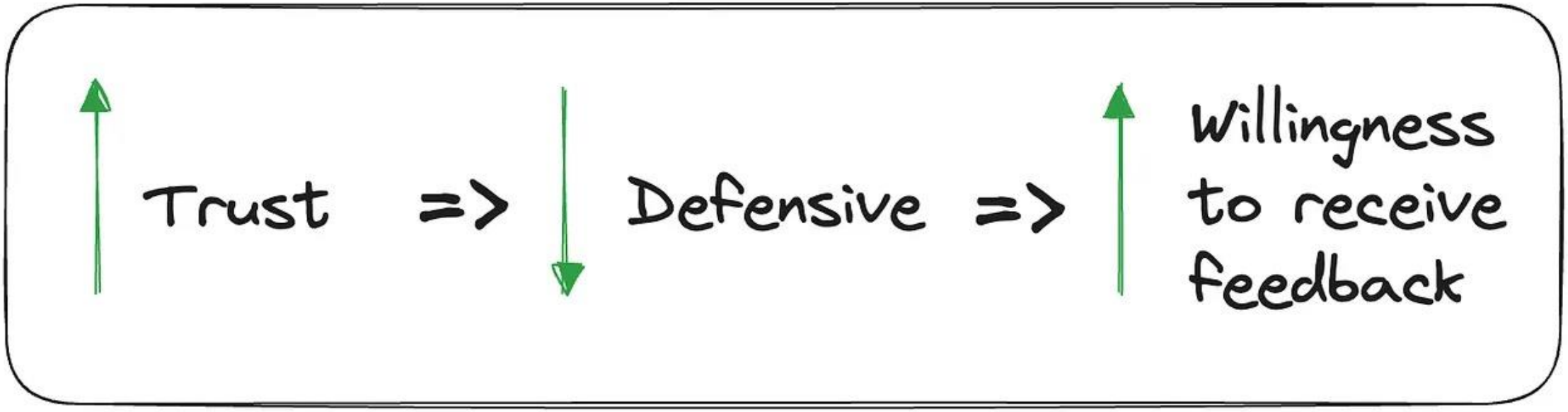
Clear communication is key

Five types of code to look for:

- Good code
- Could-be-better code
- Bad code
- “Meh” code
- Amazing code

Adopt the power of “nit”

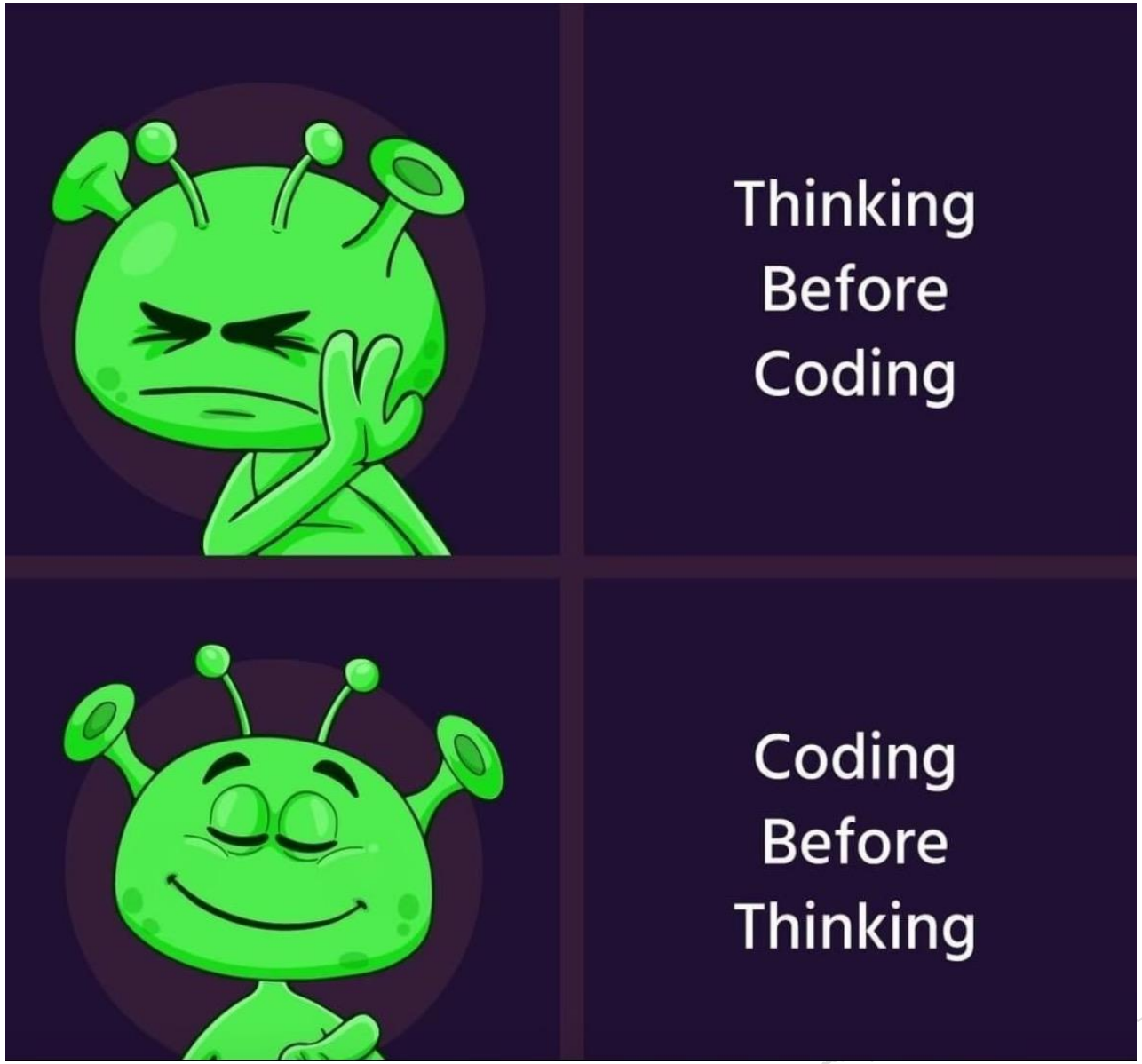
Review the Reviewer



Review the Reviewee

- Focus on design before code

Review the Reviewee



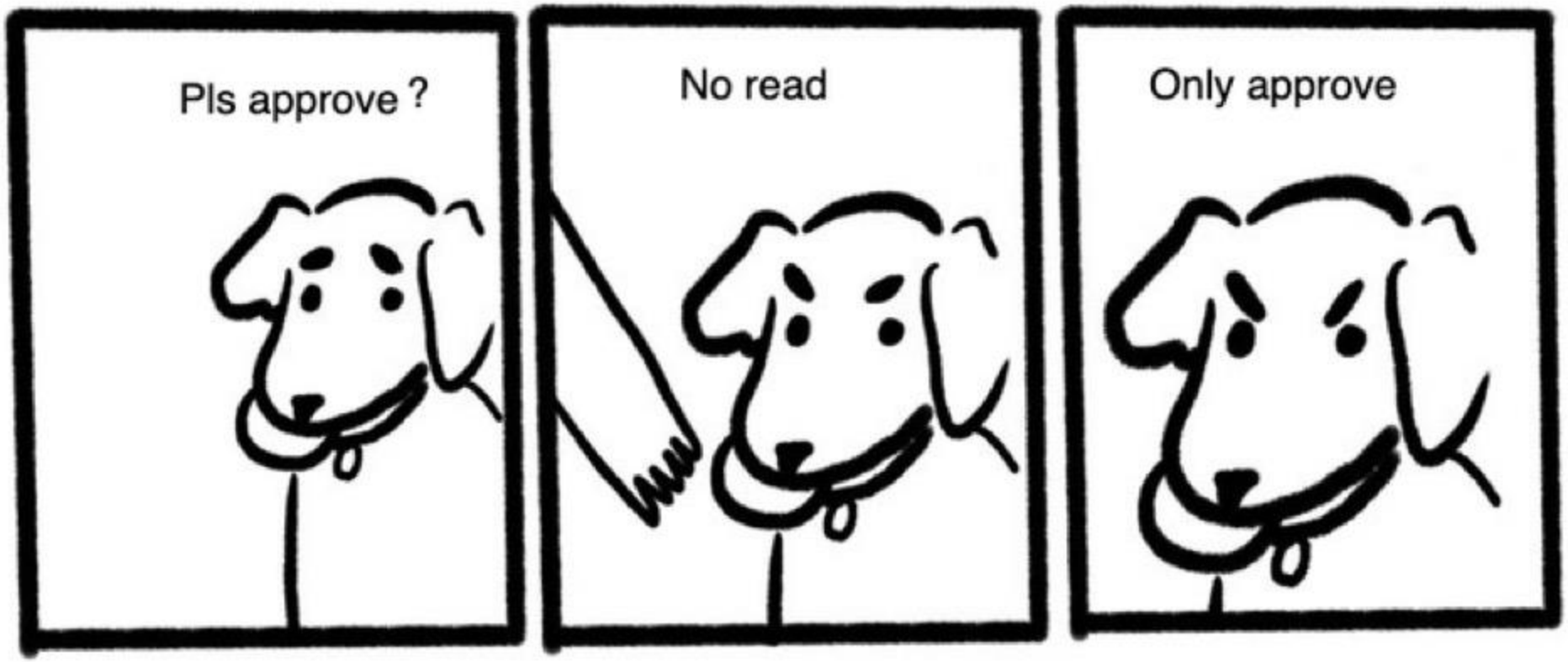
Review the Reviewee

- Focus on design first
- Review yourself
- Be proactive in clarifying unclear parts
- Be open to feedback

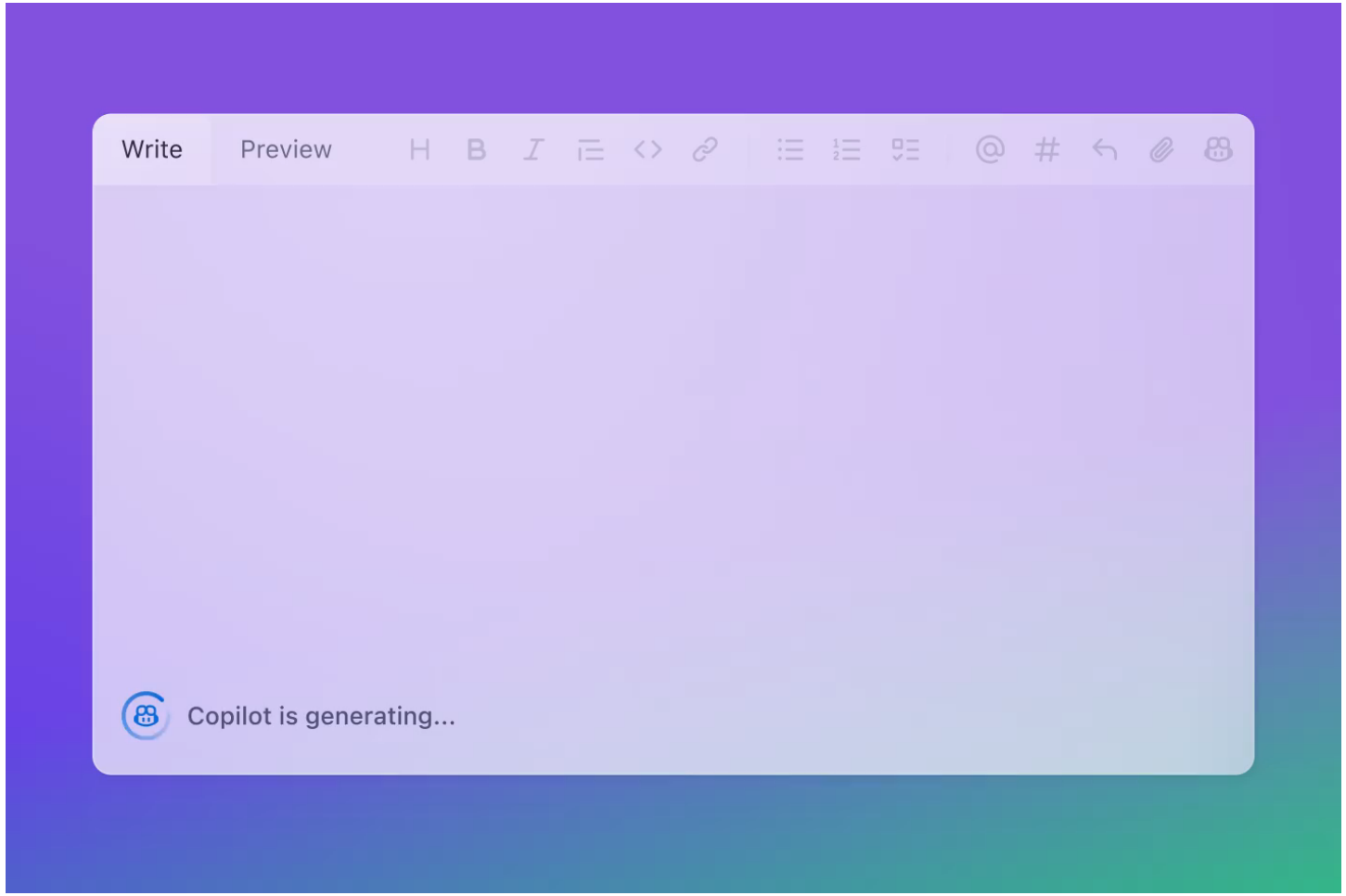
Review the Reviewee

"..I need to push this PR fast.."

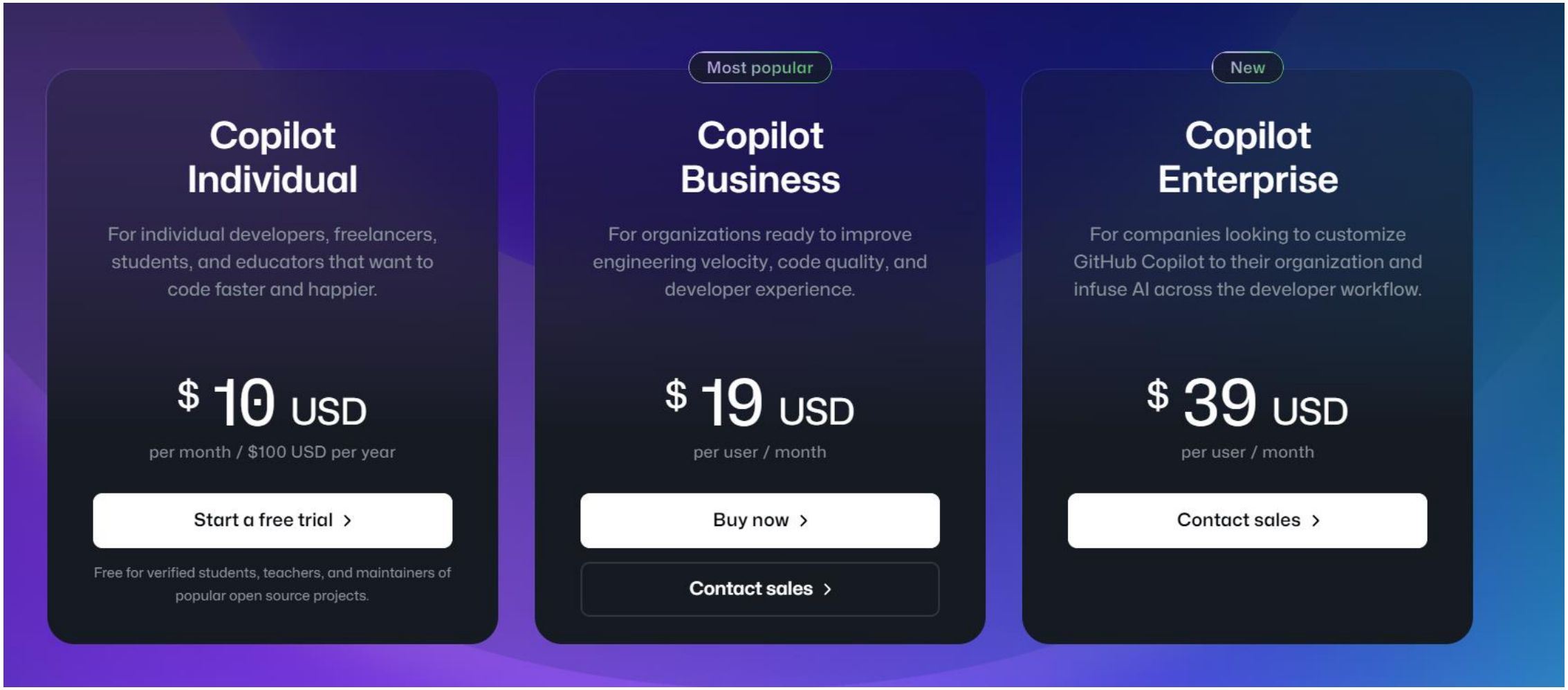
"..I don't have time to argue.."


Pls approve ?
No read
Only approve

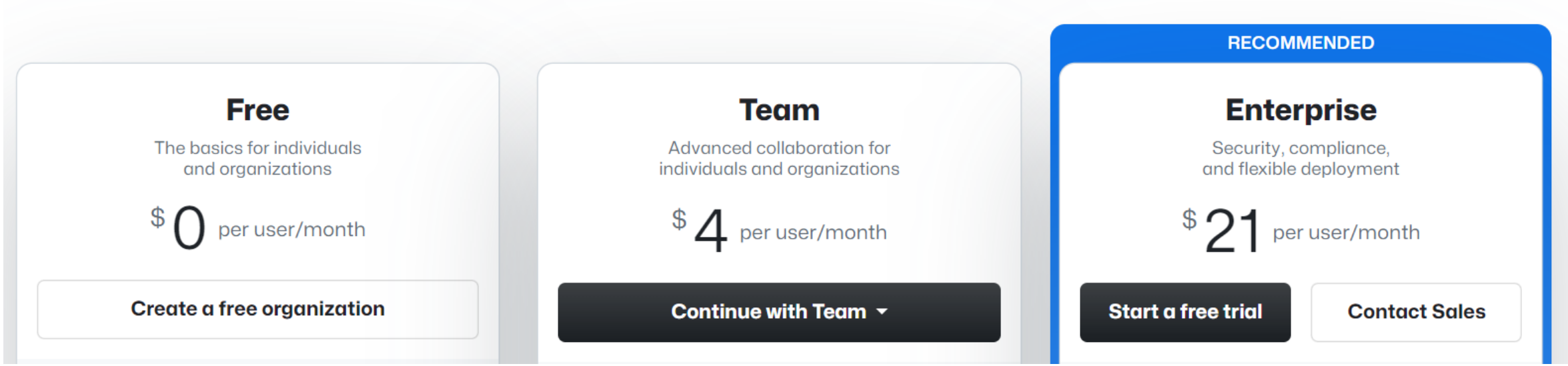
GitHub Copilot



GitHub Copilot



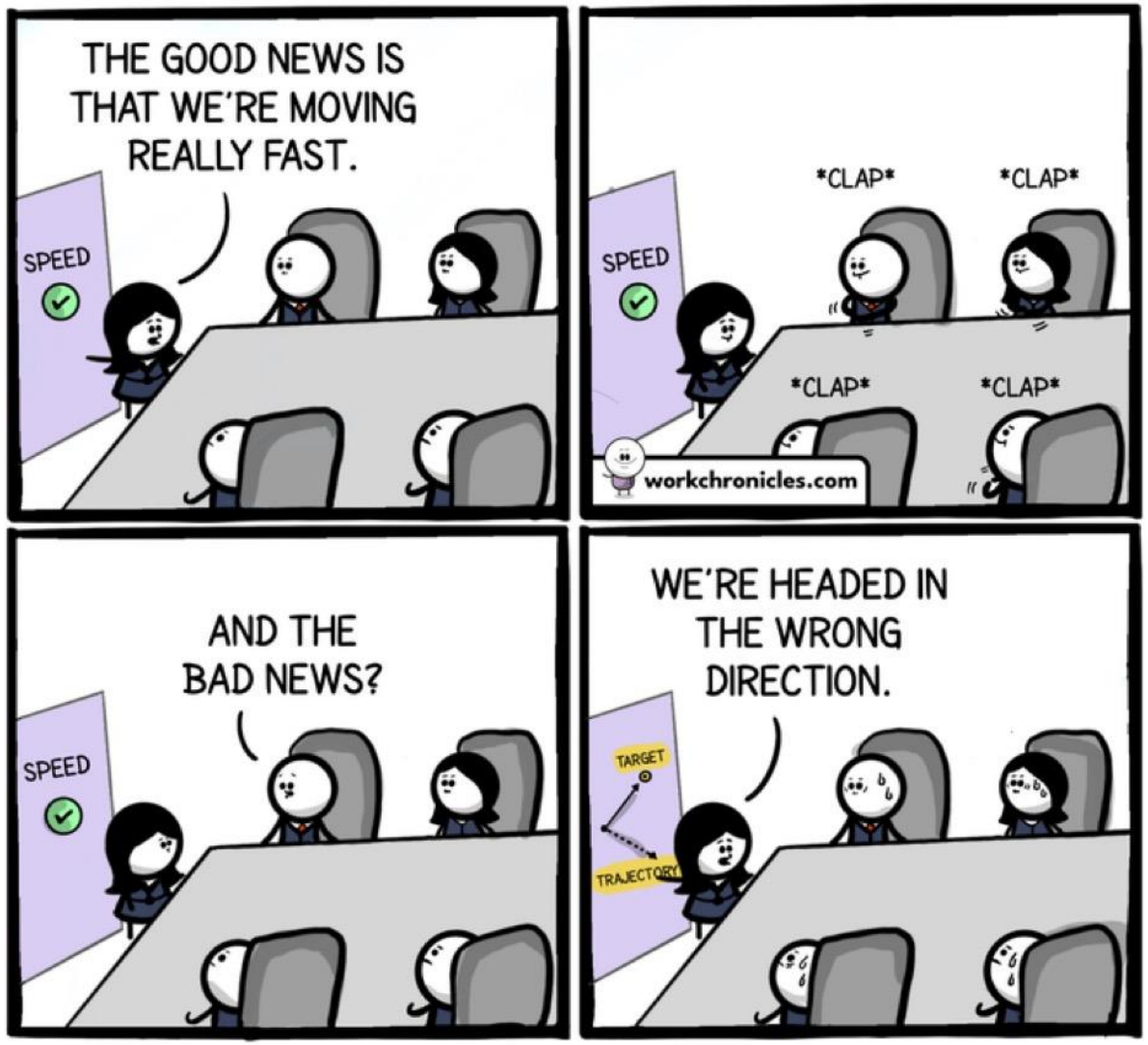
GitHub Copilot



Key Takeaways

- Do the code review

Focus on design before code

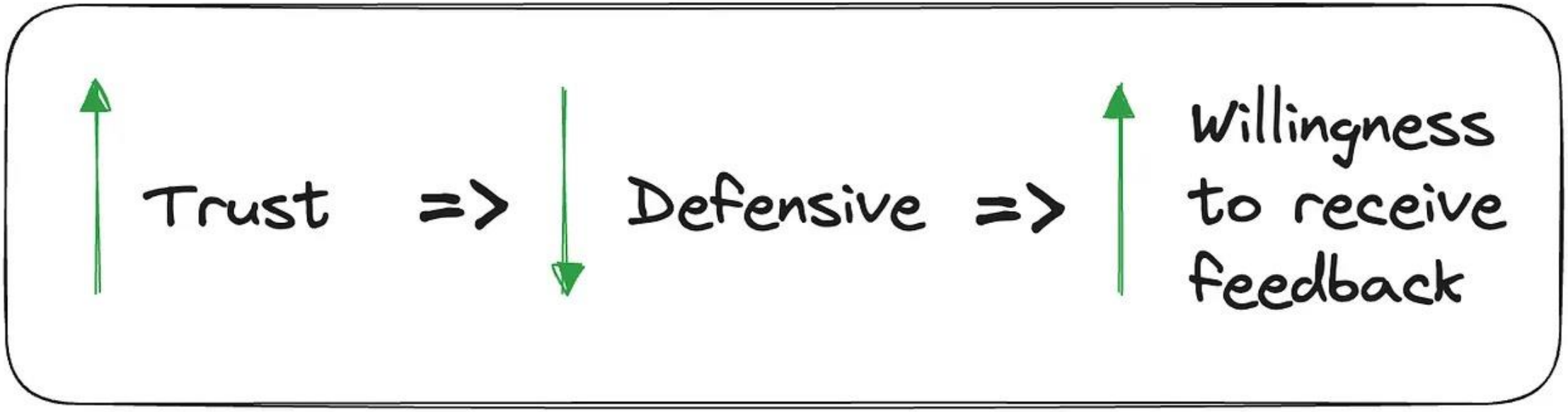


Keep pull requests small

Key Takeaways

- Do the code review
- Focus on design before code
- Keep the pull requests small
- Adopt process improvements to make reviews faster

Clearly communicate your intentions



Key Takeaways

- Do the code review
- Focus on design before code
- Keep the pull requests small
- Adopt process improvements to make reviews faster
- Clearly communicate your intention

Key Takeaways

"We need to understand that PRs aren't an obstacle that we need to overcome, but a tool to collectively have better code."

– James Pearson

BC/NAV TECH DAYS 2024
mibuso.com